



Detecting Distributed Social States from Multimodal Signals in Group Conversations

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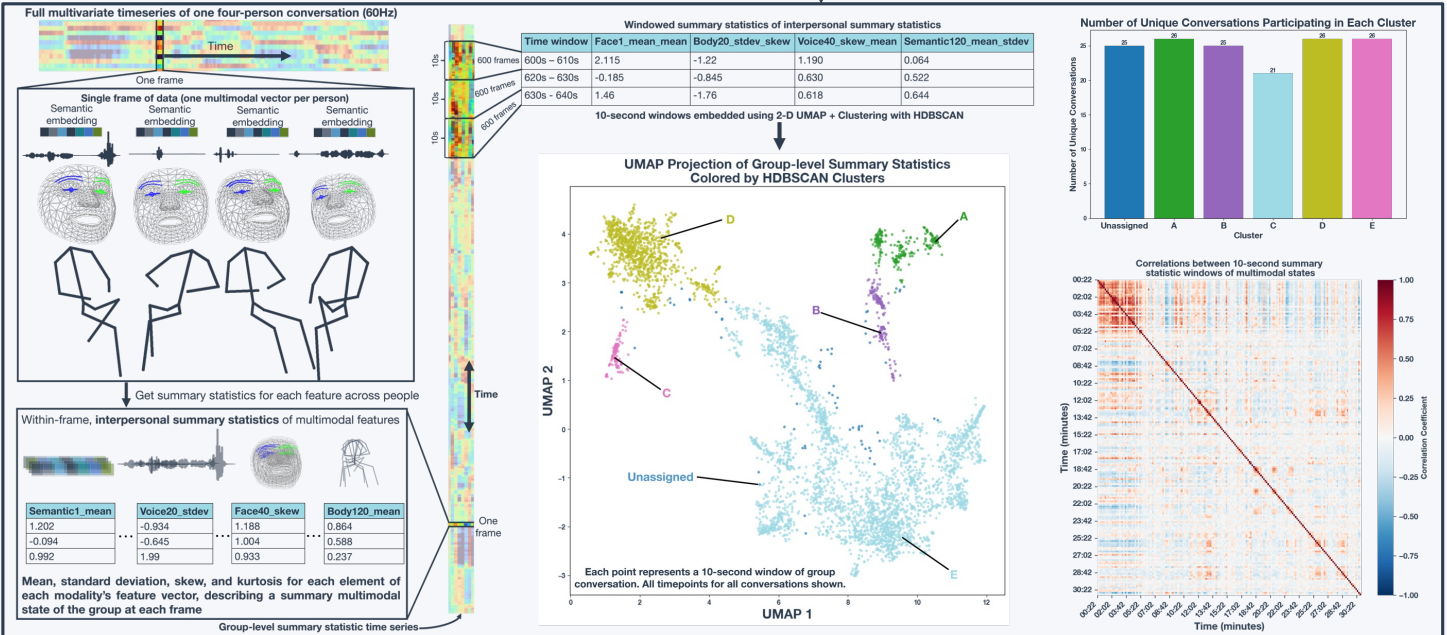
Introduction

Group conversations may transition between awkward silences, excited chatter, and heartfelt storytelling all within the span of a few minutes. This work uses data-driven methods to probe the multimodal indices of such distributed social states recurring across group conversations and understand how they correlate with participants' subjective experiences.

Methods

- Twenty-six unconstrained four-person conversations (N=104) were recorded at 30 minutes each using an array of cameras and microphones
- Pretrained deep neural network (DNN) models used to extract **facial expressions**, **3D body motion**, **vocal acoustics**, and **speech semantics** from recordings
- Collective social states were derived from aggregates of groups' conversational behaviors during **non-overlapping 10-second windows**
- Uniform manifold approximation (UMAP) and hierarchical density-based clustering (HDBSCAN) were used to discover clusters of recurrent social states
- Linear mixed-effects models used to predict participants' **post-conversation survey responses** as a function of the **proportion of time spent in each cluster**

Analysis Pipeline



Results

